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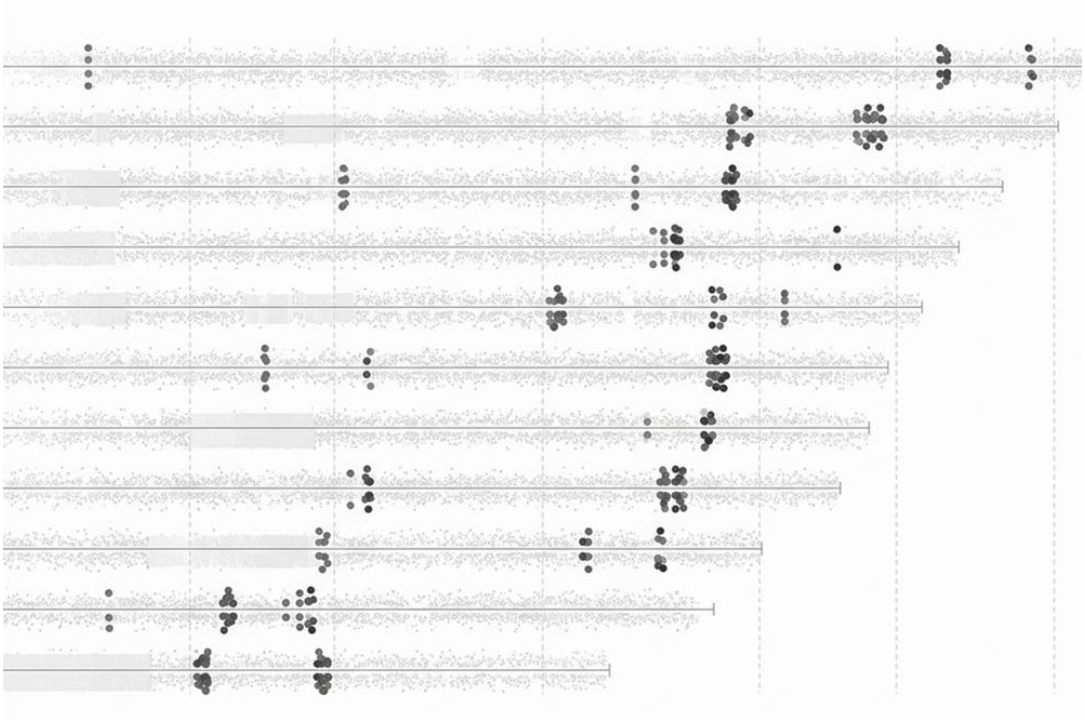
High-Resolution Variant Discovery: A Managed Workflow via Galaxy Platform

A Methodological Workflow for Targeted Genomic Analysis in Individual HG00128



BRIEFLY DESCRIPTION

This study identifies and characterizes germline variants (SNPs and INDELs) within a genomic fragment from individual HG00128 (1000 Genomes Project) using the Galaxy platform and the hg19 reference genome.

Work developed as part of my technical portfolio in Biostatistics and Omics Data Science.

Dataset: HG00128 (1000 Genomes Project) January 2026
Methodology: Genome filtering and variant calling (Galaxy)

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1. Introduction

The **1000 Genomes Project** has been fundamental in establishing a reference catalog of **human genetic variability** (1000 Genomes Project Consortium, 2010). However, it also highlights the need for specialized tools that allow for a rigorous analysis of the **massive genomic data generated by such initiatives**.

This study utilizes the **Galaxy platform** to implement an accessible **proof of concept** for genomic variant calling and filtering. While traditional high-throughput pipelines are designed for the simultaneous analysis of large cohorts, this focused approach prioritizes **methodological transparency and reproducibility**. Regardless of the reduced computational scale, this workflow establishes a rigorous framework for data reliability, ensuring the accurate detection and characterization of **Single Nucleotide Polymorphisms (SNPs)** and **small insertions/deletions (INDELs)**.

2. Objectives

- Upload FASTQ (.fq) archive and perform quality control studies.
- Process and align raw sequencing reads against the reference genome.
- Identify genetic variants, including single nucleotide polymorphisms (SNPs) and insertions/deletions (INDELs).
- Conduct a descriptive statistical analysis of the variants and their biological impact, complemented by the visualization of specific genomic fragments and variants using genome browsers.

3. Methodology

The bioinformatics analysis was conducted entirely within the **Galaxy platform (web-based platform)**. For every tool executed, specific parameter adjustments are detailed, while remaining settings were kept at their default values.

Genomic sequences were acquired in .fq format, representing paired-end reads from a single individual (HG00128) within the 1000 Genomes Project framework. For this specific study, targeted genomic fragments were selected to ensure a controlled computational environment. To ensure full reproducibility, these datasets are hosted in a dedicated GitHub repository (see Reproducibility statement).

After data upload, a **Quality Control (QC) phase was performed using FastQC**, leading to sequence refinements. The reads were then aligned to the *Homo sapiens* reference genome (hg19) using the **BWA-MEM algorithm**. The resulting BAM files underwent quality filtering, including the removal of PCR duplicates.

For the identification of SNPs and INDELs, the **FreeBayes** probabilistic variant caller was utilized. Functional annotation and biological impact prediction were subsequently performed using **SnpEff**, generating descriptive statistics on mutation distribution and typology. Variant filtering was executed with **SnpSift**, applying strict criteria to isolate 'HIGH' or 'MODERATE' impact variants. Finally, the alignments and detected variants were visually inspected using the **JBrowse** genome browser to validate the findings.

4. Results

4.1 Data acquisition & upload

The selected genomic fragments consist of **paired-end reads** from individual **HG00128**. This sequencing strategy involves reading both ends of each DNA fragment: the **forward read (R1)** and its complementary **reverse read (R2)**.

Data was uploaded to the **Galaxy platform** using the *'Get Data > Upload from disk'* tool. The file type was manually set to **fastqsanger** to ensure correct quality score encoding, while the reference genome field was kept as 'unspecified' for this initial stage.

4.2 Read Quality control (QC)

In the side panel, the **FastQC Read quality reports** tool was selected for each file separately. The default parameters were maintained without modifications.

In the report for the first file (R1), the single alert identified was a **warning regarding GC content** (Figure 1). All other quality metrics, including per-base sequence quality and per-sequence quality scores, remained within the optimal range.

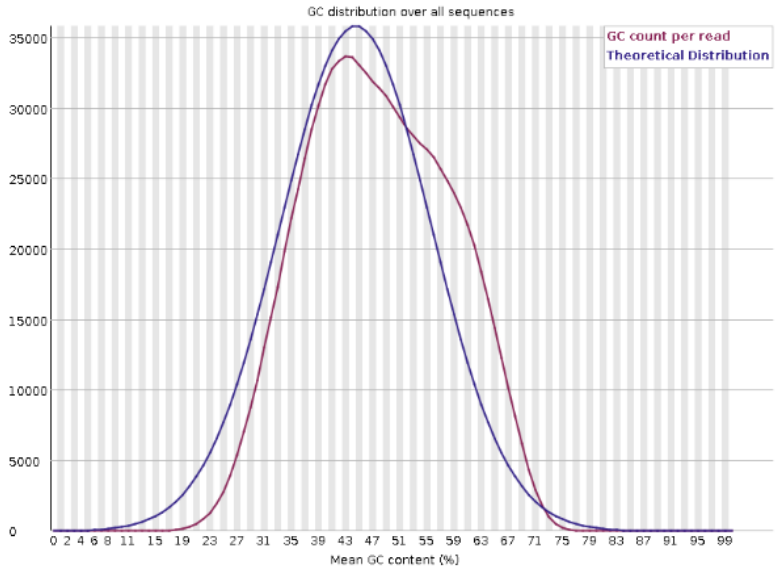


Figure 1. GC content in the forward archive.

In the second file (R2), in addition to the GC content warning, which mirrored the structure seen in Figure 1, a 'Per Tile Sequence Quality' error was identified (Figure 2).

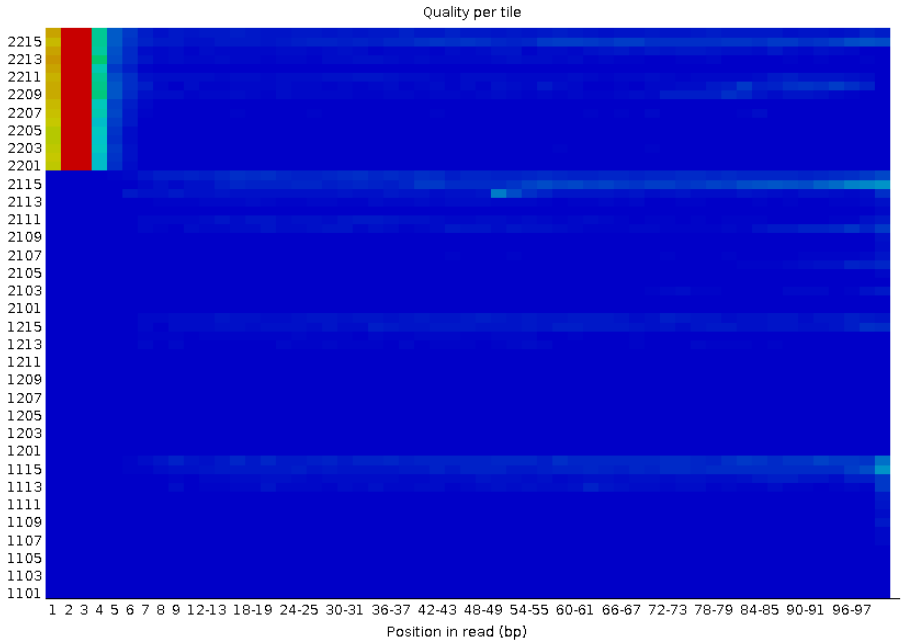


Figure 2. Per Tile Sequence Quality report for the reverse read (R2). The heatmap reveals a localized technical artifact in specific flow cell tiles, necessitating a targeted trimming strategy to maintain data integrity.

To address the low quality observed at the beginning of the reads, the **Trimmomatic tool** was employed using the HEADCROP operation. The full configuration is detailed in Appendix 1. We performed a 5-base pair crop at the start of both files, resulting in the refined sequences: R1_paired and R2_paired.

A second **FastQC** analysis was conducted on these paired sequences to verify that the previous quality issues had been resolved. The successful results of this post-trimming quality control can be found in Appendix 2 and 3.

4.3 Sequence alignment

The **Map with BWA-MEM tool** was utilized for aligning reads (optimized for sequences ≥ 100 bp). The following configuration parameters were manually adjusted, while all other settings remained at their default values:

- Reference Genome Selection: 'Use a built-in genome index'.
- Reference Genome: 'Human (Homo sapiens) (b37): hg19'.

- Read Type: 'Paired-end'.
- Input Datasets: Selected the refined R1_paired and R2_paired files.
- Read Group Information: Enabled 'Set read groups (SAM/BAM specifications)'.
- Read Group Identifier (ID): 6 (personal example)
- Read Group Sample Name (SM): sample6 (personal example)

4.3.1 Post-Alignment Refinement: PCR Duplicate Removal

Following the alignment, a single **BAM file** was generated by merging the paired-end reads based on their complementarity. This archive is available in GitHub repository (see Reproducibility statement). To ensure data integrity, **exact PCR duplicates** were identified and removed using the **MarkDuplicates** tool.

The configuration was set to *'REMOVE_DUPLICATES = true'* and *'ASSUME_SORTED = true'*. Post-execution, the generated statistics revealed that **1,565 duplicate copies** were removed, representing **0.001657%** of the total sequence information.

The output is a fully **refined BAM file**. To ensure compatibility with the downstream **FreeBayes** variant caller, the dataset attributes were manually updated to set the reference genome build to: **Human Feb. 2009 (GRCh37/hg19)**.

4.4 Genetic variant identification

With the refined BAM file ready, we proceeded to **variant calling** to identify positions that differ from the **hg19** reference genome. Among the various tools available, we selected **FreeBayes**, a Bayesian genetic variant detector designed to find small polymorphisms based on literal sequences (haplotypes) rather than individual alignments (Garrison & Marth, 2012). This approach is particularly effective for the accurate detection of **INDELs**.

The execution was performed via the Galaxy tool panel: *'FreeBayes > Choose the source of the reference genome'*. We selected the **locally cached** option and specified the **Homo sapiens hg19** build. The final output of this process is a **VCF (Variant Calling Format)** file containing the discovered genetic diversity. VCF file is available in GitHub repository (see Reproducibility statement).

4.4.1 Post-Calling Quality control

Following the execution of FreeBayes, a rigorous quality control step was performed using **SnpSift Filter**. To ensure high-confidence results, a quality threshold was applied in the *Filter criteria* field: **(QUAL >= 30)**. The use of parentheses in the expression is critical to prevent syntax errors within the tool (Cingolani et al., 2012). This filtering process resulted in a refined list of genomic variants, effectively removing

low-confidence calls originating from regions with insufficient sequencing depth or technical noise (Figure 3).

Chrom	Pos	ID	Ref	Alt	Qual	Filter
chr1	14677	.	G	A	67.5579	.
chr1	14907	.	A	G	45.2679	.
chr1	14930	.	A	G	102.107	.
chr1	762273	.	G	A	96.5961	.
chr1	762589	.	GGCC	CGCG	58.8136	.
chr1	762601	.	T	C	90.3733	.
chr1	762632	.	T	A	80.8185	.
chr1	808922	.	G	A	70.7102	.
chr1	821887	.	A	G	57.2623	.
chr1	876499	.	A	G	51.593	.
chr1	883625	.	A	G	68.0725	.
chr1	887560	.	A	C	87.0398	.

Figure 3. High-confidence variants identified by FreeBayes post-quality filtering. This refined VCF dataset exclusively includes variants meeting the (QUAL >= 30) threshold, ensuring a 99.9% confidence level for downstream functional annotation.

4.5 Variant annotation and genomic visualization

To prioritize potentially pathogenic variants, we performed a targeted filtering step using **SnpSift Filter**. The criteria were set to isolate variants with high or moderate functional impact: **(ANN[*].IMPACT = 'HIGH') | (ANN[*].IMPACT = 'MODERATE')**. No pathogenic variants were identified in this specific genomic fragment, which is consistent with the healthy population profile of the **1000 Genomes Project** (1000 Genomes Project Consortium, 2010).

To characterize the complete variant landscape, we utilized **SnpEff** with the *'Download on demand'* option for the **hg19** reference genome. The analysis identified a total of **22,526 variants** (Figure 4A). Notably, 41 of these were multi-allelic sites. Regarding variant types (Figure 4B), the dataset consists of **20,829 SNPs**, 496 MNPs, 517 insertions, and 623 deletions. This information is available in GitHub repository in .html format (see Reproducibility statement).

Functional impact analysis (Figure 4C) revealed that the majority are **modifier effects (66.876%)**, corresponding to non-coding regions. Only **0.743%** are classified as high-impact functional variants. In terms of functional class (Figure 4D), **52.848% are silent**, **46.56% are missense**, and **0.593% are nonsense** mutations. Finally, the **missense/silent ratio of 0.881**—being close to 1—confirms that these variations are not random sequencing artifacts but a reflection of human evolutionary selection, where silent mutations are naturally more prevalent.

A

Genome	hg19
Date	2026-01-18 18:14
SnpEff version	SnpEff 5.2 (build 2023-09-29 06:17), by Pablo Cingolani
Command line arguments	SnpEff -i vcf -o vcf -s snpeff_stats.html hg19 /corral4/main/objects/e/0/9/dataset_e09b4fd0-c746-4bfe-8480-293eb6b93feb.dat
Warnings	418
Errors	0
Number of lines (input file)	22,485
Number of variants (before filter)	22,526
Number of not variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	22,526
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	41
Number of effects	65,014
Genome total length	3,137,161,265
Genome effective length	3,119,100,022
Variant rate	1 variant every 138,466 bases

B

Number variants by type

Type	Total
SNP	20,829
MNP	496
INS	517
DEL	623
MIXED	61
INV	0
DUP	0
BND	0
INTERVAL	0
Total	22,526

C

Number of effects by impact

Type (alphabetical order)	Count	Percent
HIGH	483	0.743%
LOW	11,957	18.391%
MODERATE	9,095	13.989%
MODIFIER	43,479	66.876%

D

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	8,722	46.56%
NONSENSE	111	0.593%
SILENT	9,900	52.848%

Missense / Silent ratio: 0.881

Figure 4. Descriptive statistics of the genomic variants identified by FreeBayes and annotated via SnpEff. (A) Technical Summary: Overview of the annotation process and primary variant counts, confirming the successful processing of the refined dataset.(B) Variant Nature & Type: Distribution of the 22,526 variants categorized by molecular type, highlighting the prevalence of SNPs (20,829) over MNPs, insertions, and deletions.(C) Functional Impact Distribution: Quantification of variants based on their predicted biological effect, showing a dominance of Modifier (66.88%) effects in non-coding regions.(D) Functional Class Summary: Break-down of protein-coding changes, revealing the distribution between Silent (52.85%), Missense (46.56%), and Nonsense (0.59%) mutations.

Following the statistical analysis, we performed a targeted genomic visualization of the specific fragment from individual HG00128. For this task, we utilized **JBrowse**, ensuring the selection of the correct reference genome (hg19).

The visual representation includes two main distinct tracks:

- **Genomic Sequence Track**: Displaying the raw reference sequence and read coverage (see configuration in Appendix 4).
- **Variant Track Group**: Superimposing the discovered variants (SNPs and INDELs) found within this fragment (configuration in Appendix 5).

This graphical representation (Figure 5) allows for a concurrent and interactive inspection of the read alignment, the identified variations along the specific fragment, and their respective biological nature.

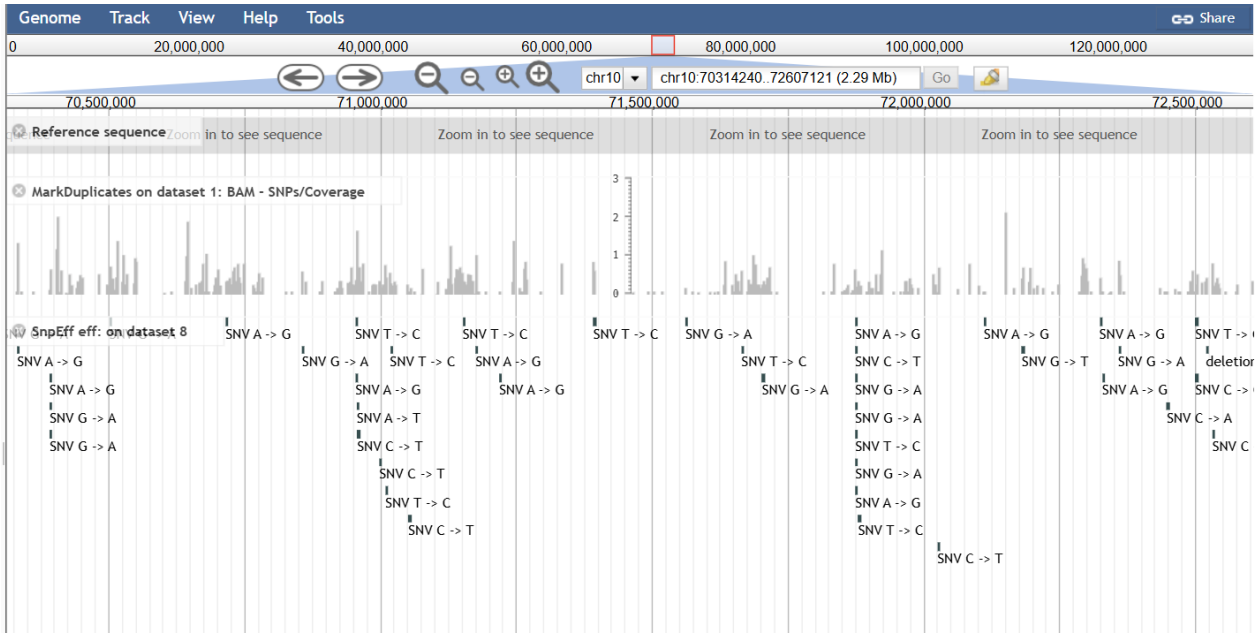


Figure 5. JBrowse visualization of a specific genomic region. The integrated view displays the reference sequence and reads alignment alongside the identified variants, allowing for a high-resolution spatial inspection of the genetic landscape.

5. Discussion

This study encompasses the essential stages for the acquisition, processing, and analysis of genomic data. The workflow began with raw sequences in .fq format. It is important to note that this is not the primary output of the sequencer, as previous steps—such as the generation of BCL files—were required to produce these much more readable, though intermediate, datasets.

Post-upload, Quality Control (QC) proved critical. Regarding the identified warnings, the GC content exhibited a slight shift from the theoretical normal distribution (Gaussian model); however, as the deviation was not significant (Figure 1), the sequences were maintained for further analysis.

Notably, a critical error was detected in the **Per Tile Sequence Quality of the second read** (R2), while the first read (R1) remained unaffected. Although these sequences are complementary, this discrepancy is explained by the R2 decay phenomenon (Schirmer et al., 2016). This effect is driven by the accumulation of noise during the second read due to reagent depletion or phasing/pre-phasing issues. To mitigate this, a trimming strategy was implemented in both reads to ensure accurate alignment.

During the alignment phase, selecting the correct reference genome was fundamental to generating the **BAM** files required for subsequent analysis. For variant calling, we utilized FreeBayes, a haplotype-based caller that enhances the detection of MNPs and INDELs (Garrison & Marth, 2012). While numerous variants were identified, a comparative analysis using alternative calling methods would be the next logical step to further validate these findings.

Functional characterization via **SnpEff** revealed a variability profile consistent with a healthy individual. The predominance of modifier variants (66.88%) is expected (Figure 4), as most human genetic variation occurs in intronic or intergenic regions. A current limitation in genomics remains the challenge of interpreting the regulatory effects of these non-coding regions on protein-coding sequences. Additionally, the high frequency of Low-impact variants (18.39%) aligns with the fact that most variations do not result in functional protein changes.

While this study focuses on a specific genomic fragment to optimize computational efficiency, it serves as a **proof of concept and guide for the necessary steps of a genomic analysis within Galaxy framework.** These support the basis for future and scalable for high-throughput clinical genomic pipelines.

In conclusion, the bioinformatic analysis of the HG00128 genomic fragment demonstrates the efficacy of the implemented Galaxy workflow for processing genomic regions and the robust identification and classification of germline variants.

6. Reproducibility statement

All raw genomic data and corresponding to HG00128 individual are publicly available through The International Genome Sample Resource under the accession number **the name accession already mentioned**. This ensures that the foundational evidence for our findings remains accessible for independent verification.

The complete analytical pipeline was performed using Galaxy in the web-based format. Relevant information and archives from this study are hosted in GitHub (https://github.com/lnungon/High-Resolution-Variant-Discovery-a-Managed-Workflow-via-Galaxy-Platform).

7. Bibliography

1000 Genomes Project Consortium. (2010). A map of human genome variation from population-scale sequencing. Nature, 467(7319), 1061–1073. https://doi.org/10.1038/nature09534

Cingolani, P., Patel, V. M., Coon, M., Nguyen, T., Land, S. J., Ruden, D. M., & Lu, X. (2012). Using Drosophila melanogaster as a model for genotoxic chem-ical mutational studies with a new program, SnpSift. Frontiers in Genetics, 3, 5. https://doi.org/10.3389/fgene.2012.00005

Garrison, E., & Marth, G. (2012). Haplotype-based variant detection from short-read sequencing. arXiv preprint arXiv:1207.3907. https://doi.org/10.48550/arXiv.1207.3907

Schirmer, M., Ijaz, U. Z., D'Amore, R., Hall, N., Sloan, W. T., & Quince, C. (2016). Insight into biases and sequencing errors for amplicon sequencing with the Illumina MiSeq platform. Nucleic Acids Research, 44(6), e57. https://doi.org/10.1093/nar/gkw018

8. Appendix

Appendix 1

Trimmomatic Configuration (Preprocessing)

To address the technical artifacts identified during the initial Quality Control, the **Trimmomatic** tool was configured with the following parameters:

- **Read Type:** *Paired-end* (utilizing two separate input files: R1 and R2).
- **ILLUMINACLIP Step:** *Enabled* (using default settings) to remove any potential adapter contamination.
- **Headcrop Operation:** A **HEADCROP** of **5 bp** was applied to the start of every read to eliminate low-quality bases at the 5' end.
- **Sliding Window:** *Enabled* (default settings) to perform a local quality scan and trim reads where the average quality drops below the threshold.
- **Minimum Length (MINLEN):** Set to **35 bp**. Any reads falling below this length after trimming were discarded to prevent non-specific mapping during alignment.

1: Trimmomatic Operation

Select Trimmomatic operation to perform

Cut the specified number of bases from the start of the read (HEADCROP)

Number of bases to remove from the start of the read *

5

2: Trimmomatic Operation

Select Trimmomatic operation to perform

Sliding window trimming (SLIDINGWINDOW)

Number of bases to average across *

4

Average quality required *

20

3: Trimmomatic Operation

Select Trimmomatic operation to perform

Drop reads below a specified length (MINLEN)

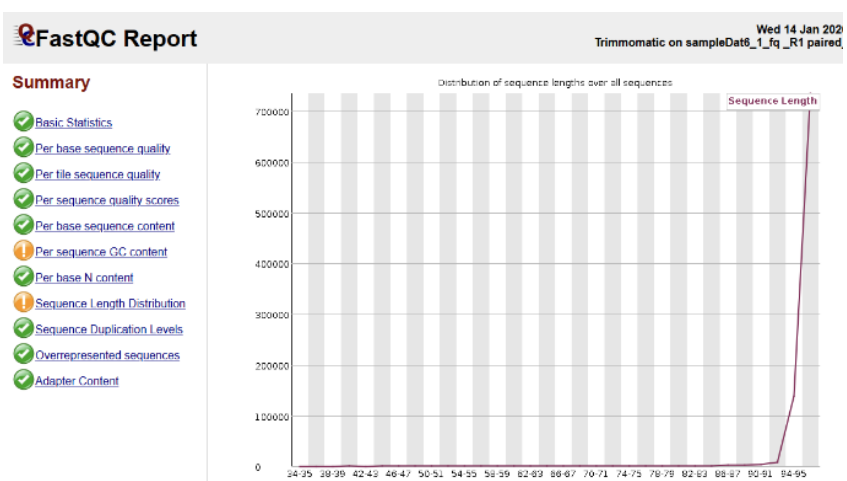
Minimum length of reads to be kept *

35

Appendix 2

Post-Trimming Quality Control (R1_paired)

While the GC content warning persists (consistent with the genomic region's profile), a new alert appears regarding Sequence Length Distribution. This warning is a direct consequence of the Trimmomatic operations, specifically the dynamic trimming based on quality thresholds. This shift from a uniform to a variable length distribution confirms the successful removal of low-quality fragments while maintaining the overall integrity of the dataset.

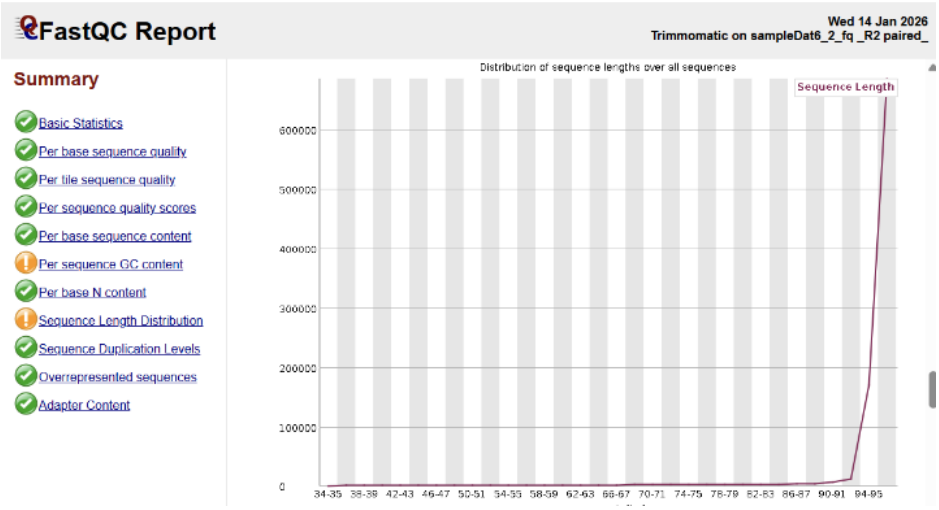


Appendix 3

Success Validation - Post-Trimming QC (R2_paired)

The most significant result is the **complete resolution of the 'Per Tile Sequence Quality' error**, which previously triggered a critical alert. By removing the initial 5 base pairs (Headcrop), the technical noise associated with the flow cell tiles was successfully eliminated.

Consistent with the forward read, the **GC content** warning persists as a biological feature of the sample, and the **Sequence Length Distribution** alert appears as an expected statistical artifact of the quality-based trimming process.



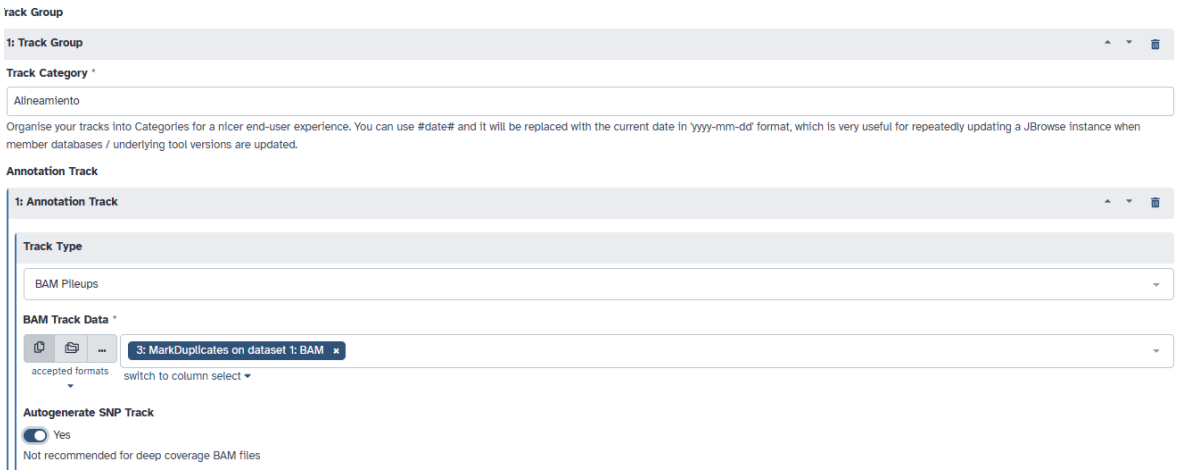
Appendix 4

JBrowse Alignment Track Configuration

To visualize the genomic alignment within the study region, a dedicated Track Group was created in JBrowse using the following parameters:

- Reference Genome: *Human Feb. 2009 (GRCh37/hg19) (hg19)*.
- Track Type: *Alignments (BAM)*.
- Input File: Refined and de-duplicated BAM file (post-MarkDuplicates).
- Autoselect Tracks: Enabled for the genomic region of interest (Individual HG00128).
- Visual Settings: Coverage histogram enabled to monitor sequencing depth across the fragment.

This configuration allows for the simultaneous inspection of individual read mappings and the overall statistical distribution of the sequencing coverage.



Appendix 5

Variant Track Configuration in JBrowse

To integrate the discovered variants into the visual analysis, a second **Track Group** was added with the following specifications:

- **Track Category:** *Variants*
- **Track Type:** *VCF SNPs*.
- **SNP Track Data:** Selected the **SnpEff annotated VCF** file (select your specific dataset).
- **Annotation Integration:** Configured to display alongside the alignment track for direct comparison between sequence reads and called variants.

This setup enables the simultaneous visualization of the reference genome, the aligned reads from individual **HG00128**, and the specific functional annotations for each polymorphism, facilitating a rapid qualitative validation of the results.